

IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Granite and LangFlow"

Project Title – NAAC Assist AI

Domain of Project – Education / Academic Compliance Automation

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Himanshu Gupta		himanshu561hi@gmail.com	+91-8090860670

Problem Statement -

Problem Statement : NAAC Assist AI

The Challenge

The current accreditation process demands an overwhelming amount of manual coordination from faculty members to gather, parse, and organize **thousands of pages of institutional data** spanning multiple academic years. Universities struggle to extract meaningful metrics and evidence from unstructured sources like policy manuals, annual reports, student logs, and committee minutes. Drafting official **Self Study Reports and NAAC Best Practices** by hand leads to inconsistencies, formatting errors, and missed compliance. Without an intelligent system, institutions suffer from **data gaps that directly result in lower accreditation grades** and loss of prestige.

The Objective

Build a **purpose-built RAG platform** that acts as a virtual accreditation assistant. The solution should:

- **Secure Document Ingestion** – Upload PDF/Word files and instantly index them locally.
- **AI Chat Interface** – Answer NAAC criteria questions anchored strictly to uploaded evidence.
- **Automated SSR Generator** – Draft Self Study Reports for all 7 official NAAC criteria using RAG.
- **Best Practices Formatter** – Restructure raw initiatives into the strict 6-part NAAC format.
- **Zero Hallucination Guarantee** – Refuse generation if evidence is absent from the database.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used **LangFlow + IBM watsonx.ai + Granite Models**

Proposed Solution -

Proposed Solution: NAAC Assist AI

The proposed solution is a **purpose-built Retrieval Augmented Generation (RAG) platform** built using **LangFlow, IBM watsonx.ai, and Granite Models** to transform raw university records into structured NAAC compliance drafts. It addresses document search, data structuring, and audit-ready report generation through a modular, privacy-first architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline. Institutional documents are locally parsed, chunked into semantic blocks, embedded using SentenceTransformers, and stored in ChromaDB — all on-premises to guarantee 100% data privacy. The most relevant evidence is retrieved and passed to IBM Granite Instruct via the Watsonx API to generate structured reports.

The system consists of three key modules:

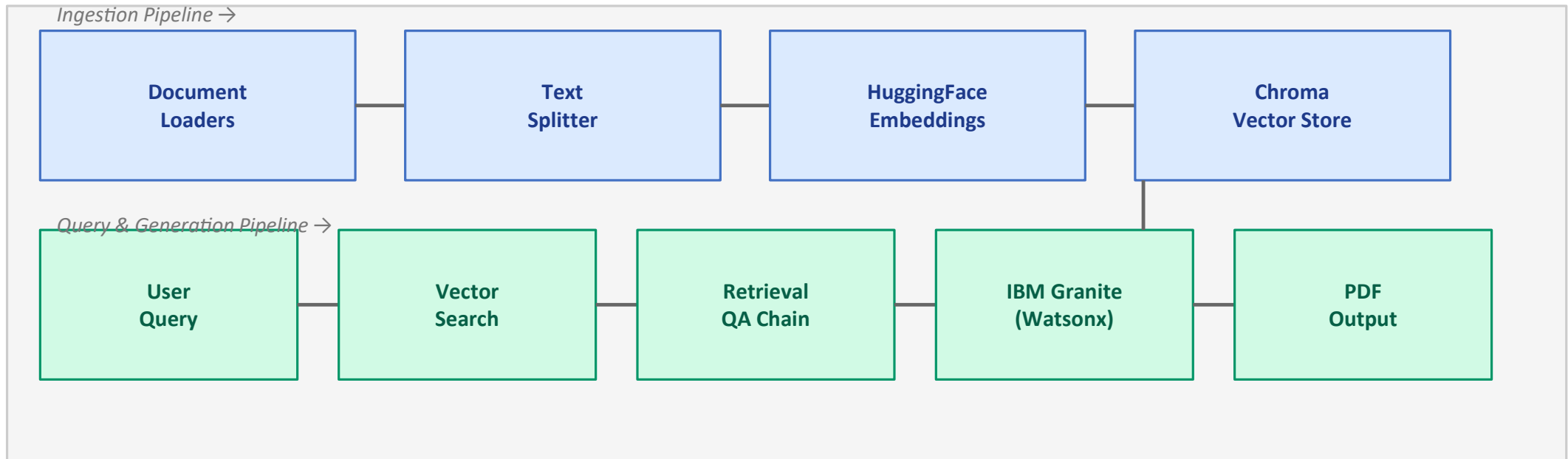
- **AI Chat Interface** allows faculty to ask freeform questions about specific NAAC criteria and receive answers strictly anchored to uploaded evidence.
- **SSR Generator Module** uses dynamic dropdowns for the official 7 NAAC criteria, instantly drafting metric responses with quantitative data.
- **Best Practices Formatter** accepts raw initiative details and restructures them into the strict 6-part NAAC format using vector-retrieved evidence.

The solution includes a **Streamlit-powered dashboard** for drag-and-drop uploads and one-click PDF export. By automating document search and report synthesis, this solution reduces preparation time from **months to minutes** while guaranteeing strict adherence to NAAC taxonomy.

Langflow component Used -

- 1) Document Loaders** – PyPDFLoader and Docx2txtLoader extract raw unstructured text from complex university documents like policy manuals and audit reports.
- 2) Recursive Character Text Splitter** – Strategically breaks large documents into smaller semantic chunks, preserving context without exceeding LLM token limits.
- 3) HuggingFace Embeddings** – SentenceTransformers (all-MiniLM-L6-v2) convert text chunks into high-dimensional vectors for semantic similarity matching.
- 4) Chroma Vector Store** – Securely indexes and persistently stores vectorized embeddings on local hardware, enabling instantaneous retrieval during queries.
- 5) Custom Prompt Templates** – Engineered with strict boundaries to ensure IBM Granite adopts an academic tone and refuses generation when context is insufficient.
- 6) Retrieval QA Chain** – Seamlessly links the vector store context and user query directly to the Watsonx LLM (IBM Granite Instruct) for NAAC report generation.

Langflow workflow -



How it works: Raw university documents → Document Loaders → Text Splitter chunks into semantic blocks → HuggingFace Embeddings converts to vectors → Chroma Vector Store saves locally. When a user submits a query, it is embedded and matched against stored vectors → Retrieved context merged with query in the NAAC Prompt Template → Watsonx LLM (IBM Granite Instruct) generates the NAAC-compliant academic response → ReportLab compiles the final formatted PDF.

Architecture blueprint

Presentation Layer	Streamlit Web Interface User Inputs Document Upload PDF Download
Application Layer	Chatbot Module SSR Generator Module Best Practice Generator Module
Data Ingestion Layer	PyPDF Parser python-docx Parser Pandas Table Extractor (runs locally)
Knowledge Base (RAG)	ChromaDB Vector Database Local Persistent Storage Similarity Search Index
AI Engine	SentenceTransformers (all-MiniLM-L6-v2) Local Embedding Generation 100% On-Premises
LLM Service + Output Layer	Watsonx API → IBM Granite Instruct ReportLab PDF Compiler

Role of Agentic AI in the solution

Role of Agentic AI in NAAC Assist AI

Agentic AI elevates this platform from a simple search engine into a highly autonomous compliance auditor. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates the entire NAAC report generation workflow autonomously — deciding which data chunks are relevant, synthesizing structured academic content, and enforcing strict generative boundaries.

The agent acts as a **virtual gatekeeper**: it refuses to generate responses or hallucinate facts if the retrieved documents do not contain sufficient supporting evidence. By **autonomously verifying the presence of quantitative data** before drafting, the AI guarantees total accuracy and reliability in Self Study Reports and Best Practice summaries.

Each request performs a specific role — retrieving evidence, synthesizing report sections, verifying data completeness — while working together through the RAG pipeline. This **multi-step autonomous approach** ensures efficient task execution and accurate results.

Agentic AI also brings **context awareness** — understanding which NAAC criterion is being addressed and retrieving only the most relevant institutional data chunks. It supports **adaptive synthesis**, continuously structuring evidence into clean academic formats based on each NAAC metric.

Overall, Agentic AI makes NAAC Assist AI proactive, evidence-anchored, and goal-driven, acting like a **smart virtual accreditation officer** that helps institutions achieve and maintain their accreditation grades.

Technology Used

- **Streamlit** – For designing and serving a secure, responsive faculty dashboard with drag-and-drop uploads and one-click PDF downloads.
- **IBM Granite Instruct (ibm-granite-13b-instruct-v2)** – Foundation model accessed via Watsonx AI for compliance-grade academic text generation.
- **IBM Watsonx.ai** – For model deployment, secure API access, embeddings governance, and enterprise-grade AI infrastructure.
- **LangFlow** – For visually orchestrating the full RAG pipeline including document loaders, text splitters, embeddings, vector store, and QA chain.
- **ChromaDB** – Local, private, persistent vector database storing all document chunks strictly on-premises — no cloud exposure.
- **SentenceTransformers (all-MiniLM-L6-v2)** – Fast, local semantic embedding generation that eliminates cloud embedding costs and data leakage risk.
- **PyPDF / python-docx / Pandas** – Robust parsing of PDFs, Word documents, and complex institutional data tables.
- **ReportLab** – Compiles LLM-generated content into beautifully formatted, academic-grade PDF documents for immediate auditor review.

Novelty and Uniqueness

NAAC Assist AI stands out by combining **local RAG, strict NAAC taxonomy enforcement, and zero-hallucination agentic AI** into a single purpose-built platform for academic institutions.

100% Local Data Privacy – All document parsing, chunking, and vector embedding happen strictly on local hardware. Sensitive university documents NEVER leave campus premises, unlike cloud-based AI tools.

Strict NAAC Taxonomy Integration – Hardcoded with the official 7 NAAC Criteria and their specific metrics, ensuring highly relevant and targeted report generation unlike generic AI wrappers.

Premium Fallback Simulation – A unique offline mode synthesizes high-quality heuristic reports using only local RAG context even when active cloud LLM credentials are unavailable.

End-to-End Automation – Manages the entire lifecycle from raw document ingestion → semantic search → instant generation → export of formatted academic PDFs with zero manual formatting.

Zero Cloud Embedding Cost – Eliminates reliance on paid cloud embedding APIs, drastically reducing operational costs while accelerating document processing via local SentenceTransformers.

Hallucination-Free Architecture – Architecturally designed to refuse generation if supporting evidence is absent — making it uniquely reliable for high-stakes accreditation submissions.

Scalable & Institution-Ready – Built on IBM watsonx.ai with IBM Granite models ensuring enterprise-grade reliability, governance, and adaptability for any accreditation framework.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)

with format –

Please test and Paste the working Github URL –

<https://github.com/himanshu561hi/faculty-NACC->

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4) any other files

.

.

Future Scope

1. Direct ERP Integration –

The system architecture can be expanded to include direct API integration with University Management Systems and ERPs to automatically pull live real-time data for student enrollment, fees, and faculty profiles — eliminating manual data entry.

2. Multi-Agent Collaboration –

Future versions will implement Multi-Agent Collaboration where distinct AI agents act as separate departmental auditors, independently reviewing and verifying each other's compliance reports for maximum accuracy.

3. Automated Evidence Cross-Checking –

The ingestion pipeline will be upgraded to instantly flag missing signatures, stamps, or dates in uploaded institutional proofs before the final audit submission.

4. Multi-Framework Support (NBA, NIRF, ABET) –

Because the architecture is highly modular, the platform can be rapidly adapted to support other major accreditation standards simply by updating the prompt taxonomy.

Thank You!

Thank you for your time and interest.